

ECE 4020-5020: Digital Signal Processing

Homework 4 (100 pts.)

Due on Tuesday, April 14, 2026

Problem 1 (20 pts.): Textbook Problem 5.18

Problem 2 (20 pts.): Textbook Problem 5.23

Problem 3 (20 pts.): Textbook Problem 5.28

Problem 4 (20 pts.): Textbook Problem 4.7

Problem 5 (20 pts.): Textbook Problem 6.12

ECE 5020 Homework 4

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1 Question 1

Example 1.1 (Question 5.18). An FIR filter is described by the difference equation:

$$y(n) = x(n) - x(n - 6)$$

- **Part A:** Compute and plot its magnitude and phase response.
- **Part B:** Determine its response to the inputs

- $x(n) = \cos\left(\frac{\pi}{10}n\right) + 3\sin\left(\frac{\pi}{3}n + \frac{\pi}{10}\right), \quad -\infty < n < \infty,$
- $x(n) = 5 + 6\cos\left(\frac{2\pi}{5}n + \frac{\pi}{2}\right), \quad -\infty < n < \infty,$

1.1 Simulations

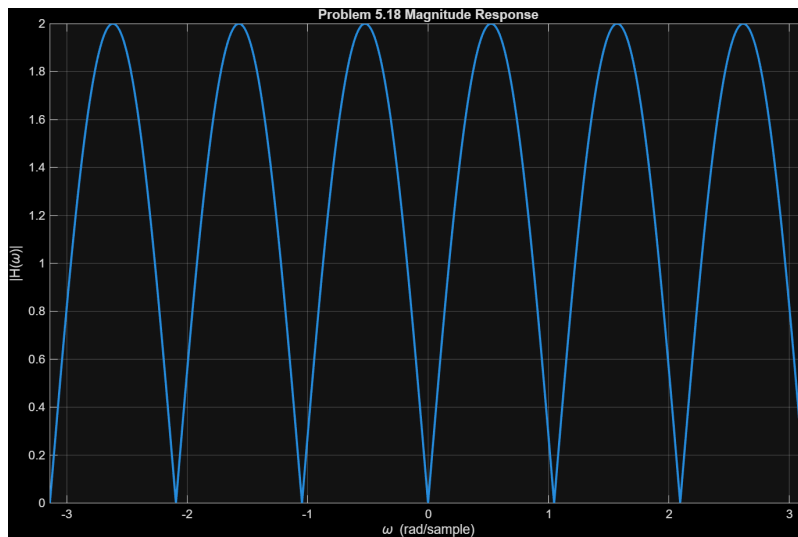


Figure 1: Magnitude Response Simulation

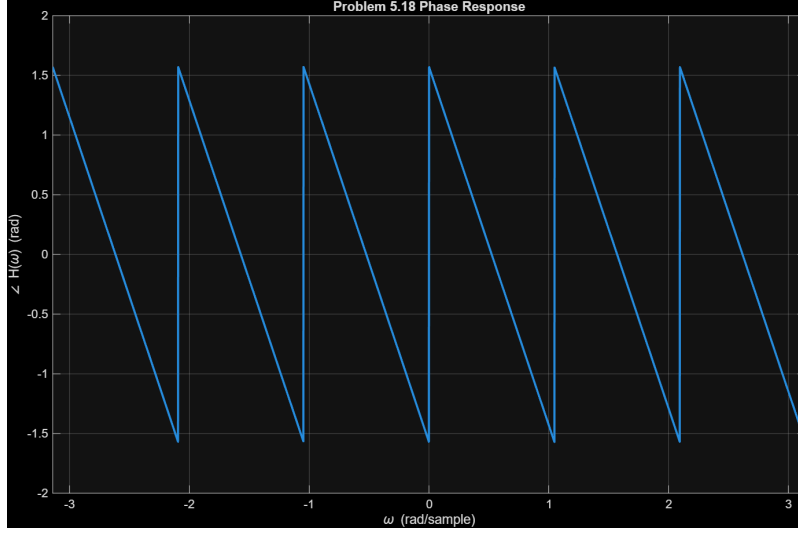


Figure 2: Phase Response Simulation

1.1.1 Part A

We are given the impulse response,

$$h(n) = \delta(n) - \delta(n - 6)$$

so it follows that the frequency response is given by:

$$H(e^{j\omega}) = 1 - e^{-j6\omega} \tag{1}$$

$$= e^{-j3\omega}(e^{j3\omega} - e^{-j3\omega}) \tag{2}$$

$$= 2je^{-j3\omega} \sin(3\omega) \tag{3}$$

Hence, it follows that we obtain:

$$|H(\omega)| = 2|\sin(3\omega)|.$$

The phase is given by the following piecewise function:

$$\angle H(\omega) = \begin{cases} -3\omega + \frac{\pi}{2}, & \sin(3\omega) > 0 \\ -3\omega - \frac{\pi}{2}, & \sin(3\omega) < 0 \end{cases}$$

Notice that the phase here is undefined at the zeros,

$$\omega = k\frac{\pi}{3}, \quad k \in \mathbb{Z}$$

Therefore, the magnitude has zeroes:

$$\omega = 0, \pm\frac{\pi}{3}, \pm\frac{2\pi}{3}, \pm\pi$$

1.1.2 Part B

Example 1.2 (Part 1). We are given:

$$x(n) = \cos\left(\frac{\pi}{10}n\right) + 3\sin\left(\frac{\pi}{3}n + \frac{\pi}{10}\right), \quad -\infty < n < \infty$$

So, for the first term $\cos\left(\frac{\pi}{10}n\right)$, we need to do the following. For $\omega_1 = \pi/10$,

$$H(e^{j\omega_1}) = 1 - e^{-j6\pi/10}$$

Its magnitude is given by:

$$|H(\omega_1)| = 2\sin\left(\frac{3\pi}{10}\right)$$

and since we know that $\sin(3\pi/10)$, the phase is:

$$\angle H(\omega_1) = -3 \cdot \frac{\pi}{10} + \frac{\pi}{2} = \frac{\pi}{5}$$

Thus, the output is given by:

$$y_1(n) = 2\sin\left(\frac{3\pi}{10}\right)\cos\left(\frac{\pi}{10}n + \frac{\pi}{5}\right)$$

Therefore, my solution is:

$$y(n) = 2\sin\left(\frac{3\pi}{10}\right)\cos\left(\frac{\pi}{10}n + \frac{\pi}{5}\right)$$

Example 1.3. We are given:

$$x(n) = 5 + 6\cos\left(\frac{2\pi}{5}n + \frac{\pi}{2}\right), \quad -\infty < n < \infty$$

So, the DC term at $\omega = 0$ gives us:

$$H(0) = 1 - 1 = 0$$

As a result, it makes the constant 5 be eliminated. Now, let's consider sinusoidal terms. For $\omega = 2\pi/5$, we get:

$$H(e^{j2\pi/5}) = 1 - e^{-j12\pi/5} = 1 - e^{-j2\pi/5}$$

and

$$|H(\omega)| = 2\sin\left(\frac{\pi}{5}\right), \quad \angle H(\omega) = -\frac{\pi}{5} + \frac{\pi}{2} = \frac{3\pi}{10}$$

Hence, we get the output:

$$y(n) = 6 \cdot 2\sin\left(\frac{\pi}{5}\right)\cos\left(\frac{2\pi}{5}n + \frac{\pi}{2} + \frac{3\pi}{10}\right)$$

Therefore, by simplification, we get our desired answer:

$$y(n) = 12\sin\left(\frac{\pi}{5}\right)\cos\left(\frac{2\pi}{5}n + \frac{4\pi}{5}\right)$$

2 Question 2

Example 2.1 (Question 5.23). A digital filter is characterized by the following properties:

1. It is highpass and has one pole and one zero.
2. The pole is at a distance $r = 0.9$ from the origin of the z -plane.
3. Constant signals do not pass through the system.

Now, we need to solve the following:

- Part A: Plot the pole-zero pattern of the filter and determine its system function $H(z)$.
- Part B: Compute the magnitude response $|H(\omega)|$ and the phase response $\angle H(\omega)$ of the filter.
- Part C: Normalize the frequency response $H(\omega)$ so that $|H(\omega)| = 1$.
- Part D: Determine the input-output relation(difference equation) of the filter in the time domain.
- Part E: Compute the output of the system if the input is

$$x(n) = 2 \cos\left(\frac{\pi}{6}n + 45^\circ\right), \quad -\infty < n < \infty$$

(You can use either algebraic or geometrical arguments.)

2.1 Simulations

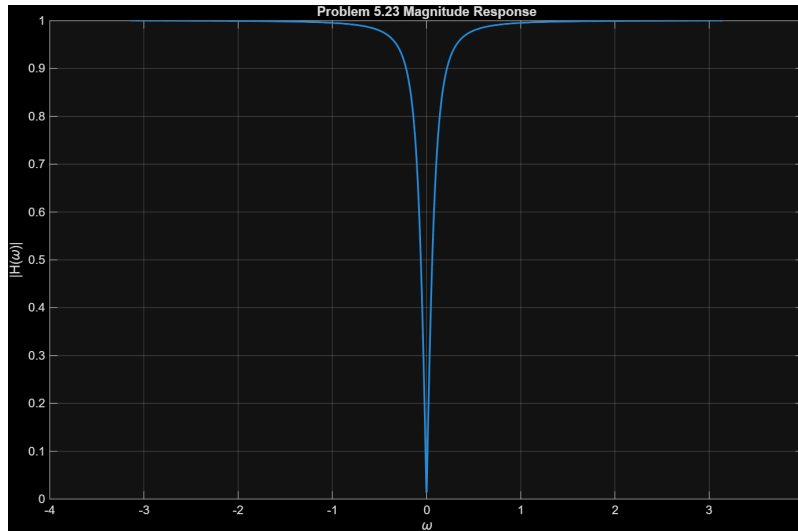


Figure 3: Magnitude Response Simulation

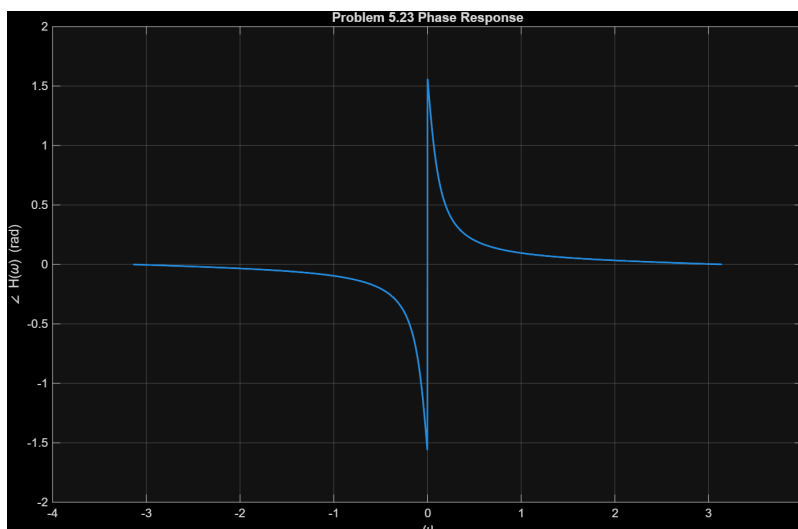


Figure 4: Phase Response Simulation

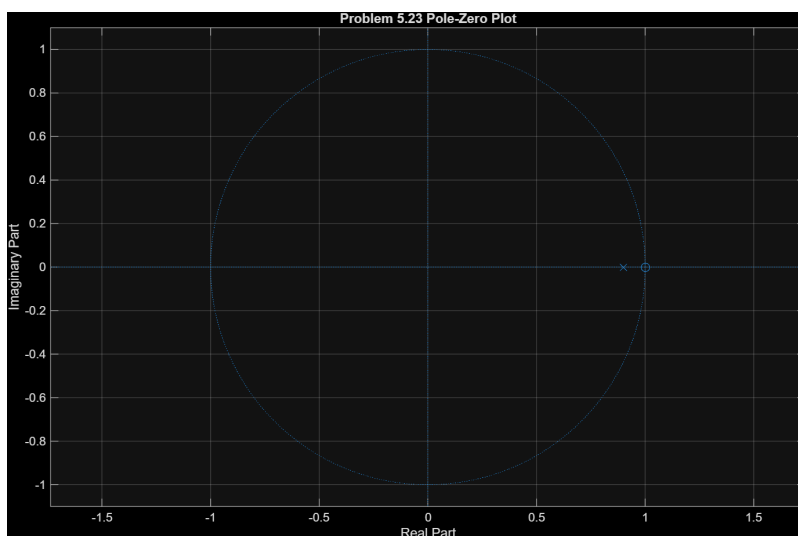


Figure 5: Simulate Pole-Zero Plot

2.1.1 Part A: Pole-Zero Pattern and System Function

So, by the definition of a *digital filter*, it follows that constant signals do not pass (i.e., $H(1) = 0$). As a result, there must be a zero at $z = 1$. For a first-order high-pass filter, the pole is placed on

the positive real axis at $z = 0.9$. Thus, the unnormalized system function is given by:

$$H(z) = \frac{1 - z^{-1}}{1 - 0.9z^{-1}}$$

where there is a zero at $z = 1$ and pole at $z = 0.9$.

2.1.2 Part B: Magnitude and Phase Response

Let $z = e^{j\omega}$. So,

$$H(\omega) = \frac{1 - e^{-j\omega}}{1 - 0.9e^{-j\omega}}$$

First, let's find the magnitude given by:

$$|H(\omega)| = \frac{|1 - e^{-j\omega}|}{|1 - 0.9e^{-j\omega}|}$$

So,

$$|1 - e^{-j\omega}| = \sqrt{(1 - \cos(\omega))^2 + \sin^2(\omega)} = \sqrt{2 - 2\cos(\omega)}$$

$$|1 - 0.9e^{-j\omega}| = \sqrt{(1 - 0.9\cos(\omega))^2 + (0.9\sin(\omega))^2} = \sqrt{1.81 - 1.8\cos(\omega)}$$

Hence,

$$|H(\omega)| = \frac{\sqrt{2 - 2\cos(\omega)}}{\sqrt{1.81 - 1.8\cos(\omega)}}$$

Next, let's do phase.

$$\angle H(\omega) = \angle(1 - e^{-j\omega}) - \angle(1 - 0.9e^{-j\omega})$$

Hence,

$$\angle H(\omega) = \tan^{-1}\left(\frac{\sin \omega}{1 - \cos \omega}\right) - \tan^{-1}\left(\frac{0.9 \sin \omega}{1 - 0.9 \cos \omega}\right)$$

An equivalent form for the first term is given by:

$$\angle(1 - e^{-j\omega}) = \frac{\pi}{2} - \frac{\omega}{2}, \quad (0 < \omega < \pi)$$

2.1.3 Part C: Normalize such that $|H(\pi)| = 1$

At $\omega = \pi$, so we get:

$$|H(\omega)| = \frac{|1 - (-1)|}{|1 - 0.9(-1)|} = \frac{2}{1.9} = \frac{20}{19}$$

So, we choose the gain to be:

$$G = \frac{19}{20}$$

Therefore, the normalized frequency response is:

$$H(z) = \frac{19}{20} \frac{1 - z^{-1}}{1 - 0.9z^{-1}}$$

2.1.4 Part D: Difference Equation

We begin from:

$$(1 - 0.9z^{-1})Y(z) = \frac{19}{20}(1 - z^{-1})X(z)$$

If we take the inverse z-transform, it follows that:

$$y(n) - 0.9y(n-1) = \frac{19}{20}x(n) - \frac{19}{20}x(n-1)$$

Therefore,

$$\boxed{y(n) - 0.9y(n-1) = 0.95x(n) - 0.95x(n-1)}$$

2.2 Part E

Given

$$x(n) = 2 \cos\left(\frac{\pi}{6}n + 45^\circ\right)$$

For an LTI system, we know that the output is,

$$y(n) = 2|H(\pi/6)| \cos\left(\frac{\pi}{6}n + 45^\circ + \angle H(\pi/6)\right)$$

Recall the previously derived normalized filter,

$$|H(\pi/6)| = \frac{19}{20} \frac{\sqrt{2 - 2\cos(\pi/6)}}{\sqrt{1.81 - 1.8\cos(\pi/6)}} \approx 0.98125$$

Thus, the output magnitude is:

$$2|H(\pi/6)| \approx 1.9625, \quad \angle H(\pi/6) \approx -11.11^\circ$$

Therefore,

$$\boxed{y(n) \approx 1.9625 \cos\left(\frac{\pi}{6}n + 33.89^\circ\right)}$$

3 Question 3

Example 3.1 (Question 5.28). A second-order system has a double pole at $p_{1,2} = 0.5$ and two zeroes at

$$z_{1,2} = e^{\pm j3\pi/4}$$

Using geometric arguments, choose the gain G of the filter so that $|H(0)| = 1$.

3.1 Solution

We need to determine $H(z)$. So,

$$H(z) = G \frac{(1 - z_1 z^{-1})(1 - z_2 z^{-1})}{(1 - 0.5z^{-1})^2}$$

At $\omega = 0$, we need to evaluate on the unit circle at $z = 1$:

$$H(0) = G \frac{(1 - z_1)(1 - z_2)}{(1 - 0.5z)^2}$$

Since we have,

$$z_1 + z_2 = 2 \cos\left(\frac{3\pi}{4}\right) = -\sqrt{2}, \quad z_1 z_2 = 1,$$

it follows that:

$$(1 - z_1)(1 - z_2) = 1 - (z_1 + z_2) + z_1 z_2 = 2 + \sqrt{2}$$

Hence,

$$|H(0)| = |G| \frac{2 + \sqrt{2}}{0.25} = 4(2 + \sqrt{2})|G|$$

Let $4(2 + \sqrt{2})|G| = 1$. So,

$$G = \frac{1}{4(2 + \sqrt{2})} = \frac{2 - \sqrt{2}}{8}$$

Therefore, the gain is

$$G = \frac{2 - \sqrt{2}}{8}$$

4 Question 4

Example 4.1 (Question 4.7). An analog signal

$$x_a(t) = \sin(480\pi t) + 3 \sin(720\pi t)$$

is sampled 600 times per second.

- Part A: Determine the Nyquist sampling rate for $x_a(t)$.
- Part B: Determine the folding frequency.
- Part C: What are the frequencies, in radians, in the resulting discrete-time signal $x(n)$?

4.1 Solution

First, we need to determine the Nyquist sampling rate for $x_a(t)$. So, the analog frequencies are given by:

$$f_1 = \frac{480\pi}{2\pi} = 240 \text{ Hz}, \quad f_2 = \frac{720\pi}{2\pi} = 360 \text{ Hz}$$

Therefore, the Nyquist rate here is:

$$F_N = 2(360) = 720 \text{ Hz}$$

The folding frequency is

$$\frac{F_s}{2} = 300 \text{ Hz}$$

Finally, we need to determine the frequencies that result in the discrete-time signal $x(n)$. So, we have

$$\omega_1 = 2\pi \frac{240}{600} = \frac{4\pi}{5}, \quad \omega_2 = 2\pi \frac{360}{600} = \frac{6\pi}{5}$$

However, the discrete-time frequency is modulo 2π , then:

$$\frac{6\pi}{5} \equiv -\frac{4\pi}{5}.$$

Hence,

$$\begin{aligned} x(n) &= \sin\left(\frac{4\pi}{5}n\right) + 3\sin\left(\frac{6\pi}{5}n\right) \\ \sin\left(\frac{6\pi}{5}n\right) &= \sin\left(2\pi n - \frac{4\pi}{5}n\right) = -\sin\left(\frac{4\pi}{5}n\right) \\ \Rightarrow x(n) &= \sin\left(\frac{4\pi}{5}n\right) - 3\sin\left(\frac{4\pi}{5}n\right) \end{aligned}$$

Then, we get

$$x(n) = -2\sin\left(\frac{4\pi}{5}n\right)$$

Therefore, the desired discrete-time frequency is

$$\omega = \frac{4\pi}{5}$$

with the second component aliasing onto the same discrete-time frequency.

5 Question 5

Example 5.1 (Question 6.12). Consider the simple signal processing system show in Fig 6.12. The sampling periods of the A/D and D/A converters are $T = 5$ ms and $T' = 1$ ms, respectively. Determine the output $y_a(t)$ of the system, if the input is

$$x_a(t) = 3 \cos 100\pi t + 2 \sin 250\pi t, \quad (t \text{ in seconds})$$

Since the postfilter removes frequencies above $F_s/2$ of the D/A side, where

$$T' = 1 \text{ ms} \Rightarrow F'_s = \frac{1}{T'} = 1000 \text{ Hz}$$

So, the post-filter cutoff is

$$\frac{F'_s}{2} = 500 \text{ Hz}$$

5.1 Solution

First, sample with $T = 5 \text{ ms}$. Since $T = 0.005 \text{ s}$,

$$x(n) = x_a(nT) = 3 \cos 100\pi T + 2 \sin 250\pi T$$

Then, with $T = 0.005 \text{ s}$, so

$$100\pi T = \frac{\pi}{2}, \quad 250\pi T = \frac{5\pi}{4}$$

Hence,

$$x(n) = 3 \cos\left(\frac{\pi}{2}n\right) + 2 \sin\left(\frac{5\pi}{4}n\right)$$

Now, since

$$\begin{aligned} \frac{5\pi}{4} &\equiv -\frac{3\pi}{4}, \quad (\text{mod } 2\pi) \\ \Rightarrow \sin\left(\frac{5\pi}{4}n\right) &= -\sin\left(\frac{3\pi}{4}n\right) \end{aligned}$$

Therefore,

$$x(n) = 3 \cos\left(\frac{\pi}{2}n\right) - 2 \sin\left(\frac{3\pi}{4}n\right).$$

Now, we need to reconstruct with D/A period $T' = 1 \text{ ms}$. With the same discrete-time frequencies map to analog frequencies using $T' = 0.001 \text{ s}$. So, first term is given by

$$\omega_1 = \frac{\pi}{2} \Rightarrow \Omega_1 = \frac{\omega_1}{T'} = \frac{\pi/2}{0.001} = 500\pi \text{ rad/S} \Rightarrow 3 \cos(500\pi t)$$

which corresponds to 250 Hz. Finally, we need to deal with the second term:

$$\omega_2 = \frac{3\pi}{4} \Rightarrow \Omega_2 = \frac{3\pi/4}{0.001} = 750\pi \text{ rad/S}$$

Thus, it follows that

$$-2 \sin(750\pi t)$$

and this corresponds to 375 Hz. Both are below the post-filter cutoff at 500 Hz and so they pass. Therefore,

$$y_a(t) = 3 \cos(500\pi t) - 2 \sin(750\pi t)$$